

Chapter 6

Hemodynamic Attenuation of Attention Heads

6.1 Introduction

The previous chapter used blood flow principles to constrain the macro architecture of a model, at the level of cross-attention between measurement sites. This chapter applies brain blood flow principles to the micro architecture, at the level of individual attention heads. The way the brain decides which vessels to keep is used to decide which heads a transformer keeps.

In Chapter 4, a trained transformer carried its causal computation in two of forty heads, but this was found only after training, by ablating each head in turn, and the number of heads to keep had to be chosen by hand. This chapter asks whether a model can find the number of heads it needs during training, without being given it. We introduce a rule for this, hemodynamic attenuation, and test it on a task whose minimal number of heads is known.

6.2 Background

6.2.1 Blood supply

Four properties of cerebral blood supply motivate the rule:

1. Supply follows need: local blood flow rises with neural activity, signalled in part by astrocytes [2]. Each head is given a valve $g_h \in [0, 1]$.
2. A vessel is dispensable when collateral vessels can supply its tissue [3]. A head is valued by what is lost when it closes and the other heads compensate, its collateral value.
3. Maintaining a vessel has a metabolic cost [4]. Each open head has a fixed price, and there is no target number of heads.
4. Unneeded vessels regress gradually [5, 6], and collaterals can enlarge when a route is blocked [3]. Closing heads fade out over 100 steps and can be reopened.

The biology motivates the design, and no claim is made about the brain.

6.2.2 Astrocytes and attention

Softmax attention turns a query’s scores $s_1, \dots, s_N$ into weights

$$a_i = \frac{e^{s_i}}{\sum_j e^{s_j}}, \quad (6.1)$$

a numerator that differs for each memory slot and a denominator that is one number shared by all of them. Kozachkov et al. [1] showed that a network of neurons and one astrocyte can approximately compute self-attention. The neurons store the numerators in Hebbian synapses. The astrocyte computes the denominator: it pools activity across all the synapses it wraps and scales every one of them by the same gain, $1/p_t$. For token t the output is approximately (their Eq. 11)

$$\ell_t \approx \frac{1}{p_t} \sum_i e^{k_i^\top q_t} v_i + x_t, \quad p_t = \sum_j e^{k_j^\top q_t}, \quad (6.2)$$

which is softmax attention plus a residual connection. The point used in this chapter is that the astrocyte acts as a single gain on a whole block of synapses.

6.2.3 Head pruning

Michel et al. [10] remove heads by a gradient score, and Voita et al. [11] learn head gates with an L_0 penalty. Optimal Brain Damage [7] scores a parameter by the loss increase when it is removed, and Optimal Brain Surgeon [8] by the increase after the remaining parameters are re-fitted. ZipLM [9] applies the latter to whole heads of trained language models. All of these prune after training, to a size chosen by the user.

6.3 Methods

6.3.1 Task

The memory has $N = 16$ slots, each holding a random item c_j . A query specifies a position p , and the target is the items at R fixed offsets (Figure 6.1a):

$$\text{target}(p) = (c_{p+\delta_1}, \dots, c_{p+\delta_R}), \quad \text{slots modulo } N. \quad (6.3)$$

Items are redrawn for every example, so the model must learn where to attend.

A head returns, for each query, one weighted average of the items: one linear equation in the R unknown items. Recovering R items needs R independent equations, so the task needs $k^* = R$ heads, and with $k < R$ heads the lowest achievable loss is

$$\text{loss}(k) = \left(1 - \frac{k}{R}\right) L_{\text{triv}}, \quad (6.4)$$

where L_{triv} is the loss of predicting the mean. Duplicate heads give the same equation and count once. Measured losses agree with Equation (6.4) to within 1% (Figure 6.1c).

The task counts as solved when the loss is below $0.02 L_{\text{triv}}$. Trained models sit hundreds of times below this threshold, and losing one needed head raises the loss by about L_{triv}/R , so results are unchanged for any threshold from 0.01 to 0.1.

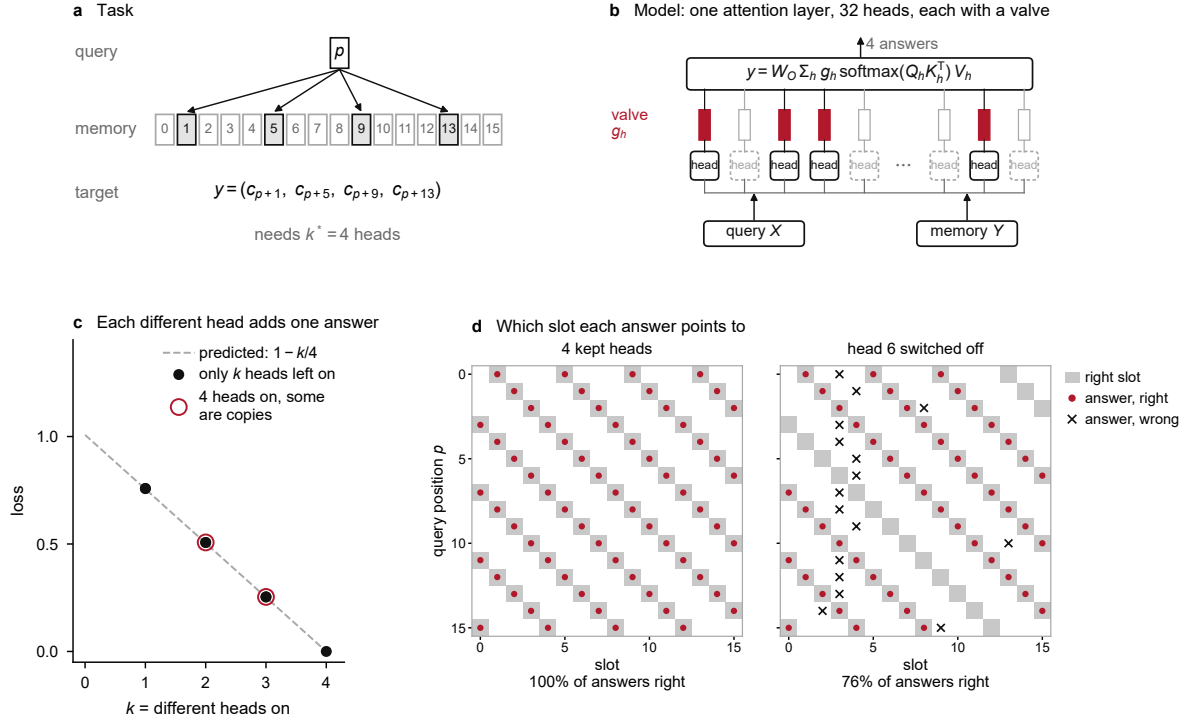


Figure 6.1: Task and model. **a**, A query carries a position p , and the target is the items stored 1, 5, 9, 13 slots ahead, so 4 heads are needed. **b**, One attention layer with 32 heads, each with a valve g_h . Filled valves are open. **c**, Loss with k different heads on a trained 4-head model, against the prediction $1 - k/4$. **d**, The slot each answer points to: all correct with the 4 kept heads, 76% with head 6 switched off.

6.3.2 Model

The model is one cross-attention layer with $H = 32$ heads and no MLP. Matrices have one column per token, as in [1]: X holds the queries and Y the memory. Head h computes the usual attention,

$$O_h = V_h \text{softmax}\left(\frac{K_h^\top Q_h}{\sqrt{d_k}}\right), \quad Q_h = W_Q^h X, \quad K_h = W_K^h Y, \quad V_h = W_V^h Y, \quad (6.5)$$

with the softmax taken down each column, so each query's weights over the memory add up to one. Each head's output is multiplied by its valve g_h before the heads are added together:

$$\hat{T} = \sum_{h=1}^H g_h W_O^h O_h. \quad (6.6)$$

A head with $g_h = 0$ contributes nothing and receives no gradient, so a model with k open valves is exactly a k -head model.

The valve plays the same algebraic role as the astrocyte gain: one number that multiplies a whole block of synapses, here the output weights W_O^h of one head. The two differ in what sets them and how fast they change (Table 6.1). The correspondence is algebraic only.

	Astrocyte gain [1]	Head valve
One number for	all synapses of one astrocyte	all output weights of one head
Set by	pooled input activity	collateral value relative to a price
Changes	with every token	slowly, over training
Function	softmax denominator	selection of heads

Table 6.1: Two gains on blocks of synapses.

6.3.3 Hemodynamic attenuation

All valves start open. After a warm-up of 10% of training, every 25 steps the rule (1) computes the collateral value of every open head, (2) closes the lowest-valued head if its value is below the price $0.03 L_{\text{triv}}$, and (3) reopens the highest-valued closed head if its value exceeds twice the price. A closing valve decreases by 1/100 per step.

The collateral value is the increase in the best-fit loss when head h is removed and the other open heads re-fit their output weights:

$$v_h = \min_W L(\text{without } h) - \min_W L(\text{all open heads}). \quad (6.7)$$

This is the Optimal Brain Surgeon score [8], applied to whole heads as in ZipLM [9]. The score is not new. The contribution is its use during training, with a price, a gradual fade, and reopening. It is computed in closed form from one matrix inverse (Section 6.6).

The re-fit is what makes a copy worthless. If two heads output the same signal and the weights are (0.5, 0.5), removing one without re-fitting halves the output, so both heads look important. With re-fitting the other head takes weight 1, the loss does not change, and the copy is worth nothing.

6.3.4 Baselines and evaluation

All methods share the model, task, training and runs. Published methods, run inside our training: ZipLM-style [9] (the same score applied once, then fine-tuning), Michel et al. [10], and Voita et al. [11]. Controls remove one part of the rule: no fade, no re-fit (the Optimal Brain Damage score [7]), and no score (random choice).

We report exact count (open heads over the last 40% of training equal k^*) and never broke (after first falling below the solved threshold, the loss never exceeds it). Experiments use $k^* = 2, 4, 8$ with 10 seeds each, plus runs with duplicated heads, runs with random head failure, and a two-layer induction task [12].

6.4 Results

Count and stability. Hemodynamic attenuation kept exactly k^* heads in 29 of 30 runs and never broke the model in 30 of 30. No other method achieved both (Table 6.2). Without the fade the model broke in every run. Without the re-fit it kept too many heads. With random choice it broke in 24 of 30 runs.

Duplicated heads. With every head starting as a copy of another, the rule kept 4 heads and never both members of a pair. Without the re-fit, 8 to 15 heads and 4 to 7 pairs were kept (Figure 6.2d).

Method (30 runs each)	Exact count	Never broke
Hemodynamic attenuation	29/30	30/30
ZipLM-style [9]	24/30	0/30
Michel et al. [10]	28/30	0/30
Voita et al. [11]	9/30	0/30
No fade	30/30	0/30
No re-fit	2/30	30/30
No score (random)	30/30	6/30

Table 6.2: Exact count and stability on the retrieval task ($k^* = 2, 4, 8, 10$ seeds each).

Backup heads. When heads fail at random, the rule should keep extra heads in proportion to how likely they are to be needed. A formula written before the experiments (Section 6.6) predicted the number kept exactly in 21 of 24 runs and to within one head in the rest (Figure 6.2e).

Induction. In a two-layer attention-only model with $8 + 8$ heads, trained to predict the token that followed an earlier copy of the current token, the minimal circuit is one previous-token head and one induction head [12]. Hemodynamic attenuation kept $2 + 1$ heads in 10 of 10 runs, one more than the minimum, and never broke the model. The kept heads had the expected roles in all 10 runs, and switching off the induction head reduced next-token accuracy on the repeat to zero (Figure 6.3). On this task every method kept the same $2 + 1$ heads, so only stability separated them: the method of Michel et al. broke the model in all 10 runs, while the controls without re-fit or with random choice did not.

Cost. Final loss was 2 to 7 times the dense model’s, well below the threshold, using 0.31 to 0.45 of its head-steps. The saving is nominal, since closed heads are still computed.

6.5 Discussion

A rule built from how vessels are maintained and pruned recovered a known minimal circuit during a single training run, without being given its size and without the model leaving the solved regime. Removing any one of its components lost either the count or stability.

The tasks are small and the main model has one layer, which is what makes the answer known, but the results do not yet extend to large language models. The collateral value is exact only for one layer, and the compute saving requires skipping closed heads in code. The biology motivates the rule but is not evidence for it.

Two experiments would connect this chapter to the rest of the thesis and have not yet been run: applying the rule to qSA-Cross, to test whether it keeps the two causal heads found in Chapter 4, and to the Cross-Site Transformer on WildPPG, to test whether the same heads are kept across all sixteen leave-one-subject-out folds.

6.6 Technical details

Collateral value in closed form. Let Z hold the outputs of the open heads on a batch (n columns) and T the targets. The best output weights are $W^* = TZ^\top(ZZ^\top)^{-1}$. With

$M = (ZZ^\top/n)^{-1}$ and J the rows of head h ,

$$v_h = \frac{1}{d_{\text{out}}} \text{tr}(W_J^* (M_{JJ})^{-1} W_J^{*\top}), \quad (6.8)$$

so one inverse serves all heads. This added about 5% to training compute. In one layer $ZZ^\top/n$ is the exact curvature of the loss in the output weights. Deeper models would need a layer-wise approximation [9].

Backup heads. If each head fails with probability p at every step, with B heads open and R needed, a head's expected value is

$$v(B) = \frac{1-p}{R} \Pr [\text{Binomial}(B-1, 1-p) \leq R-1], \quad (6.9)$$

the probability that it works times the probability that fewer than R of the others do. The rule should keep the largest B with $v(B)$ above the price.

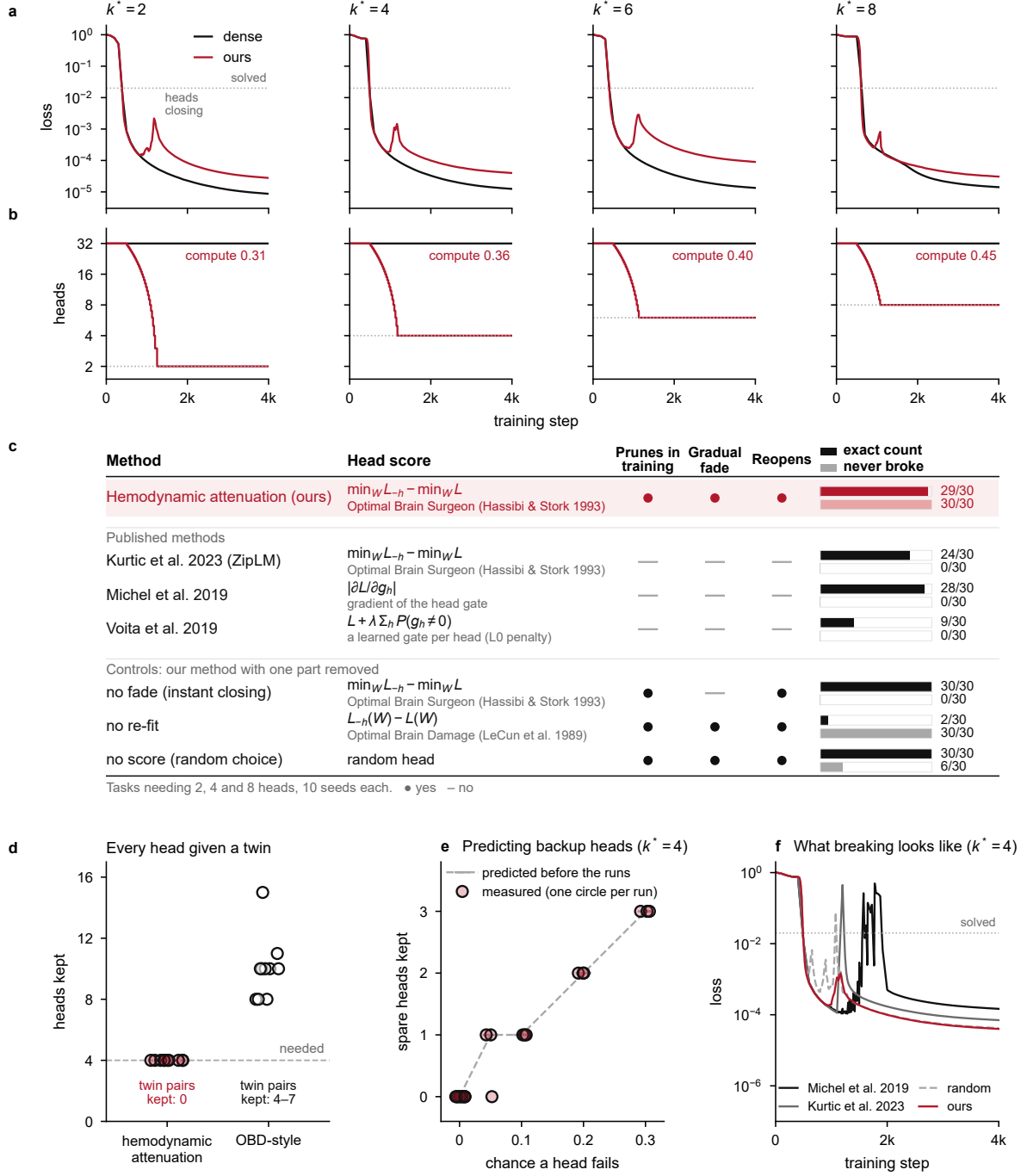


Figure 6.2: Hemodynamic attenuation during training. **a**, Loss of the dense model (black) and with hemodynamic attenuation (red). **b**, Open heads, with k^* dotted. **c**, All methods on the same 30 runs. **d**, Runs in which every head starts with an identical copy. **e**, Spare heads kept under random failure, against the prediction. **f**, One run per method. Only hemodynamic attenuation stays below the solved threshold.

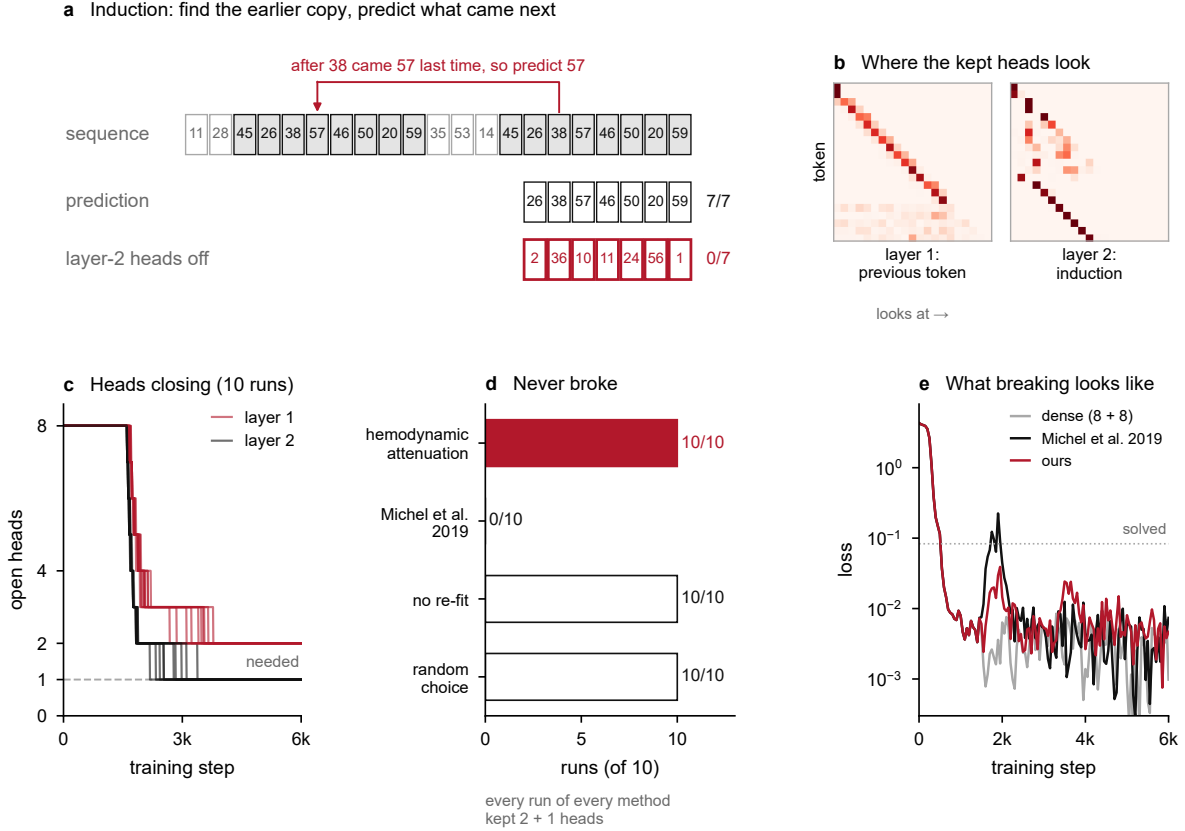


Figure 6.3: Two-layer induction. **a**, An example sequence and the model’s predictions on the repeat, with all kept heads (7/7 correct) and with the layer-2 head switched off (0/7). **b**, Attention of the kept heads on that sequence (rows: each token, columns: where it looks). The layer-1 head looks one token back, and the layer-2 head looks at the token after the earlier copy. **c**, Open heads in each layer during training, 10 runs. Layer 2 reaches the needed one head, and layer 1 stops at two. **d**, Runs that never broke, 10 per method, all with the same settings. Every run kept 2 + 1 heads. **e**, Loss during training for one seed.

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